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Persistence of harmful algal blooms under conditions of internal phosphorus loading

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ABSTRACT

Phosphorus release from sediments in lakes can trigger harmful algal blooms, significantly impacting lake ecosystems and necessitating effective management strategies. This study presents a mathematical analysis of a dynamic model incorporating such internal phosphorus loadings and their impact on algal growth. The model accounts for both light and phosphorus limitations on cell growth, as well as the seasonal variability of temperature and light, leading to a periodically forced non-linear system of ordinary differential equations. Using the theory of periodic semiflows and Floquet multipliers, we establish both necessary and sufficient conditions for long-term survival of algae (uniform persistence). This approach provides a threshold result for algae survival and the existence of a non-trivial periodic solution. Through numerical simulations, we illustrate our results and provide insights into the role of internal phosphorus loadings. In particular, our simulations illustrate that an integrated strategy reducing both watershed inflows and sediment phosphorus outperforms measures that focus on just one source of phosphorus.

1. Introduction

The frequency of harmful algal blooms (HABs) in lake ecosystems is increasing worldwide together with water pollution and climate change [1,2]. These blooms degrade water quality, produce toxins harmful to humans and wildlife, and impose significant economic costs [3]. A range of monitoring and control methods—such as toxin assays, sensor networks, aeration, chemical treatments, and biological agents—has been tested [4], yet each approach faces practical limits in cost, coverage, or unintended impacts. These challenges highlight the critical need for robust nutrient-reduction programs complemented by effective in-lake management techniques.

A key factor controlling algal growth is the availability of dissolved inorganic nutrients, particularly phosphorus, which is widely recognized as the ultimate limiting macronutrient in freshwater ecosystems [5]. In response, many management strategies have focused on reducing external phosphorus loading from catchments to control lake eutrophication [6]. However, these measures have not always yielded the expected improvements in water quality. A major reason for this is the role of internal phosphorus loading, whereby phosphorus accumulated in lake sediments is released back into the water column, sustaining algal blooms even after external inputs are reduced [7,8]. This phenomenon has puzzled lake managers and motivated research into strategies targeting

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sediment phosphorus, such as enhancing sorption capacity via chemical additions or hypolimnetic oxygenation, and direct sediment removal through dredging [9]. Yet, the long-term effectiveness of these interventions remains uncertain.

Mathematical modeling offers a valuable framework to assess the potential impact of such management strategies and to deepen mechanistic understanding of bloom dynamics. Several comprehensive lake ecosystem models integrate multiple processes, including external nutrient inputs, nutrient cycling, and sediment phosphorus release, [10–12], and a few have explicitly focused on phosphorus release from sediments [13,14]. Complementing these large-scale approaches, a variety of low-dimensional phytoplankton growth models have enabled formal mathematical analyses of key processes such as seasonal temperature forcing [15], algal–zooplankton interactions [16], nutrient recirculation loops [17], co-limitation by light and nutrients [18], and classical light-limited growth with photoinhibition [19]. While these models represent simplified scenarios, they serve as a theoretical simplification of complex ecosystems, providing a fundamental framework for identifying essential factors that influence algal blooms [20]. However, to our knowledge, no mathematical analysis has considered the internal loading of nutrients. Indeed, even numerical works have barely focused on the impact of sediment release [21].

In this work, we develop and analyze a novel dynamic model for algal growth that explicitly incorporates internal phosphorus release, balancing ecological realism and mathematical tractability. As a basis for our study, we consider the periodically forced light-limited Droop model studied in [22]. This model allows periodic variations in any parameters, thus allowing the different seasonal changes in environmental factors such as light intensity, temperature, and nutrient loads. Moreover, this model accounts for colimitation by light and nutrients as done by [18]. Internal nutrient loading is then incorporated by introducing a new state variable accounting for nutrient retention within lake sediments. This extension is based on several models describing phosphorus dynamics in lakes [23–25], which introduce parameters representing P release and sedimentation rates [26]. The model constitutes a four-dimensional system of ordinary differential equations (ODEs). We aim to establish necessary and sufficient conditions for the survival of algae (uniform persistence), offering a critical parameter known as basic reproduction number [27], which determines whether the algal population will go extinct or survive in the long-term.

A key feature of the model is the presence of periodic forcing, which represents seasonal changes in environmental conditions, leading to a periodic system. To study periodic systems, the Theory of Periodic Semiflows [28,29] and the Theory of Persistence [30] bring several results providing sufficient conditions for uniform persistence and existence of periodic solutions, and have been applied to study several population models based on periodic systems of ODEs. For example, epidemiological models [31–33], chemostat models [22,34–37], and even harmful algal models [38]. Additionally, the theory of asymptotically periodic semiflows shows to be specially useful for establishing threshold levels on persistence [22,36]. Finally, Floquet multipliers [39] have been used to analyze periodic chemostat models [40], but the difficulty of determining these multipliers analytically limits their use.

The purpose of this study is therefore twofold: (i) to derive and analyze a mechanistic algal growth model that integrates internal phosphorus loading under seasonal forcing, and (ii) to rigorously characterize the conditions for bloom persistence, thereby providing theoretical information for effective management strategies that mitigate HABs.

The outline of the paper is as follows. In Section 2, we introduce the mathematical model and its main assumptions. In Section 3, we present the main result related to the persistence and the existence of periodic solutions. In Section 4, we illustrate our theoretical results with numerical simulations, highlighting the impact of periodic forcing and internal phosphorus loading on the persistence of algae. Finally, we include two appendices: Appendix A presents asymptotic results on periodic Kolmogorov equations, and Appendix B presents results on the dynamics of a non-homogeneous periodic linear system that describes phosphorus dynamics in the absence of algae.

2. Model description

We consider a chemostat type model to describe the dynamics of algae and phosphorus concentration in a lake (Fig. 1). We represent algal abundance by the variable N (cells/m^3), which we assume to be uniformly distributed throughout the medium. Contrary to classical chemostat models, we assume that nutrients can accumulate at the bottom of the chemostat. As mentioned in the introduction, this assumption is inspired by shallow lakes where there is an exchange of nutrients between the water column and the sediment bed. We use the variables P_l and P_s to denote phosphorus concentration (gP/m^3) in the water column and in the sediment bed, respectively. Following classical models for algal growth [18,22] and models accounting for nutrient retention [26], the dynamics of algae and nutrients is described by the following system of ODEs:

$$\begin{aligned}
\frac{dN}{dt} &= [\mu(t,N,Q) - D(t) - m(t)]N, \\
\frac{dQ}{dt} &= \rho(t,Q,P_l) - \mu(t,N,Q)Q, \\
\frac{dP_l}{dt} &= W_l(t) - D(t)P_l + \alpha(t)P_s - \beta(t)P_l - \rho(t,Q,P_l)N, \\
\frac{dP_s}{dt} &= W_s(t) - \alpha(t)P_s + \beta(t)P_l.
\end{aligned} \tag{1}$$

The variable Q (gP/cell) represents the internal cell quota of phosphorus. The parameter D (day^{-1}) represents the dilution rate (the inverse of the hydraulic retention time), m (day^{-1}) is the mortality rate of algae, ρ (gP/day/cell) is the phosphorus uptake by algae, and μ (day^{-1}) is the specific growth rate of algae depending on light and phosphorus. The release and sedimentation rates of

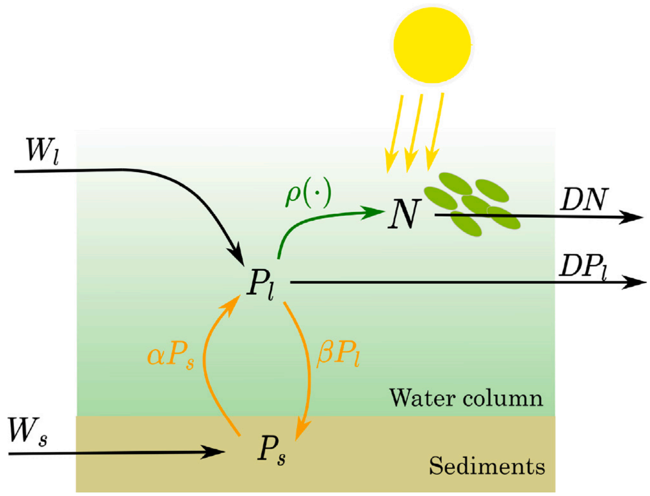

Fig. 1. Schematic representation of the algal growth model (1). External phosphorus inputs enter the available pool P_l via watershed inflow (W_l) and surface runoff (W_s); internal loading is supplied by sediment release (αP_s); sedimentation removes phosphorus at rate βP_l. Algal biomass N grows by uptaking available phosphorus through $\rho(\cdot)$ and is reduced by dilution D.

phosphorus are determined by the parameters α and β, respectively. The terms αP_s and βP_l represent the rates at which phosphorus is released into the water and removed from it through sedimentation, respectively. The parameters W_s and W_l represent phosphorus input rates to the water and sediment pool, respectively (g P/m^3/day). This model neglects grazing of algae, which is reasonable for inedible, nutritionally poor or toxic cyanobacteria [1,41].

Model parameters and functions depend on time due the impact of environmental factors such as temperature and light. The growth rate of algae is strongly dependent on light and temperature, while the coefficients associated to release and sedimentation of phosphorus depend on temperature. Taking into account the influence of temperature and light is crucial, as they can significantly reduce the growth of algae during the winter season. As a result, any potential summer algae bloom will diminish in winter due to environmental factors rather than predation.

We consider the following assumptions:

Assumption 1. All functions involved in the model, namely $\mu(t, N, Q)$, $D(t)$, $m(t)$, $\rho(t, Q, P_l)$, $W_l(t)$, $W_s(t)$, $\alpha(t)$, and $\beta(t)$, are continuous and ω-periodic with respect to time t, and take non-negative values.

Assumption 2. There are real numbers $Q^{min} > 0$ (subsistence quota) and $Q^{max} > 0$ such that $\mu(t, N, Q^{min}) = 0$ and $\rho(t, Q^{max}, P_l) = 0$ for all $t, N, P_l \geq 0$.

Assumption 3. The functions $\mu : \mathbb{R}_+ \times \mathbb{R}_+ \times [Q^{\min}, Q^{\max}] \to \mathbb{R}_+$ and $\rho : \mathbb{R}_+ \times \times [Q^{\min}, Q^{\max}] \times \mathbb{R}_+ \to \mathbb{R}_+$ are continuous in time $t \in [0, \omega]$, and locally Lipschitz with respect to the state variables, uniformly for all $t \in [0, \omega]$. Moreover, $\mu(t, N, Q)$ is decreasing in N and increasing in Q, while $\rho(t, Q, P_l)$ is decreasing in Q and increasing in P_l.

Assumption 4. There is a time t^* such that $\mu(t^*, 0, Q)$ is strictly decreasing in Q.

We note that Assumption 1 requires all model functions to be ω-periodic. This assumption is grounded in the observation that many external environmental drivers, such as temperature and light availability, exhibit seasonal patterns. Although in reality these variables do not follow perfectly periodic trajectories due to inter-annual variability and stochastic fluctuations [42], imposing periodicity is a common and practical modeling approach that facilitates rigorous theoretical analysis of models like (1). Indeed, similar periodicity assumptions have been widely adopted in phytoplankton modeling studies to represent temperature dynamics [16,43] and other seasonal forcing effects. Assumption 2 states the existence of subsistence and maximum quotas. The subsistence quota, Q^{min}, represents the value of Q at which growth ceases, while the maximum quota Q^{max} represents the quota at which absorption ceases [44]. Assumption 3 states some technical and classical assumptions on the specific growth and absorption rate functions. In particular, the decrease of μ as a function of N corresponds to self-shading. Finally, Assumption 4 states that in the absence of self-shading there is a moment of the year in which phytoplankton is limited by phosphorus (e.g. in summer). This assumption allows the application of Proposition 7 in Appendix A.

2.1. Rate functions

To better illustrate the relationship between the main model and the problem of algal blooms, as well as to illustrate our assumptions, we provide specific forms to the specific rate functions. However, it is important to note that specific models for these functions are not necessary to prove the main result (Theorem 3) of this work, which is based on the general assumptions stated above (Assumptions 1–4).

The specific growth rate of algae can be described by

$$\mu(t,N,Q) = \phi(T(t)) \min\left\{\mu_I(t,N), \mu_P(Q)\right\},$$

where ϕ is a function accounting for temperature effects, and μ_I and μ_P are functions describing the specific growth rate under limitation by light and phosphorus, respectively. These functions are given by [18]

$$\mu_I(t,N) = \frac{\mu_{max}}{(kN+K_{bg})L} \ln\left(\frac{H_I + I_{in}(t)}{H_I + I_{out}(t,N)}\right),$$

$$\mu_P(Q) = \mu_{max}\left(1 - \frac{Q^{min}}{Q}\right),$$

where μ_{max} is the maximum specific growth rate, k is the light attenuation coefficient associated to algae, K_{bg} is the background turbidity, K_I is the light half-saturation constant, L the depth of the lake, Q^{min} is the subsistence quota, $I_{in}(t)$ is the incident light intensity (generally a periodic function), and $I_{out}(t,N) = I_{in}(t)e^{-kNL-K_{bg}L}$ is the light intensity at the bottom of the lake. Properties of μ_I are found in Appendix C in [22]. The function ϕ can be described using the Cardinal Temperatures Model (CTM) [45]. When $T < T_{min}$ or $T > T_{max}$, $\phi(T)$ is defined as zero; otherwise, $\phi(T)$ is defined as:

$$\phi(T) = \frac{(T-T_{\max})(T-T_{\min})^2}{(T_{\text{opt}}-T_{\min})\left((T_{\text{opt}}-T_{\min})(T-T_{\text{opt}}) - (T_{\text{opt}}-T_{max})(T_{\text{opt}}+T_{\min}-2T)\right)}.$$

The phosphorus uptake rate can be chosen as

$$\rho(P_l,Q) = v_{max}\frac{P_l}{H_P+P_l}\left(\frac{Q^{max}-Q}{Q^{max}-Q^{min}}\right),$$

where v_{max} is the maximum absorption rate, H_P is the phosphorus half-saturation constant, and Q^{max} is the hypothetical maximum quota.

Following [26], α and β can be taken as:

$$\alpha(t) = \theta b_F(1+t_F)^{T(t)-20} \quad \text{and} \quad \beta(t) = \frac{b_S}{L}(1+t_S)^{T(t)-20} \tag{2}$$

Note that in contrast to the authors in [26], we consider a scalar $\theta \in [0,1]$ in the definition of α. The parameter θ represents the application of a method for controlling internal loads, for example, hypolimnetic oxygenation [9] or the application of chemicals such as Phoslock [46]. The introduction of the parameter θ allows to evaluate the importance of internal loadings in simulations.

Finally, the following descriptions for W_s and W_l are proposed by [26],

$$W_s = D(1-\delta)P_{in} \quad \text{and} \quad W_l = D\delta P_{in},$$

where P_{in} is the external input concentration, and δ is the fraction of P_{in} entering the lake water pool (the rest enters the sediment pool) and is defined by

$$\delta = \frac{\sqrt{365D}}{1+\sqrt{365D}}.$$

3. Mathematical analysis

In this section, we work under Assumptions 1–4 and focus on solutions that remain biologically meaningful. Specifically, we consider initial conditions belonging to the set:

$$X := \{(N,Q,P_s,P_l) \in \mathbb{R}^4 : N,P_s,P_l \geq 0, Q^{min} \leq Q \leq Q^{max}\}. \tag{3}$$

As we prove later (Lemma 4), X is positively invariant.

3.1. Absence of algae

We begin the mathematical analysis characterizing the behavior of (1) when $N = 0$ (*i.e.* the case free of algae), that is, we study the following system:

$$\begin{aligned} \frac{dQ}{dt} &= \rho(t,Q,P_l) - \mu(t,0,Q)Q, \\ \frac{dP_l}{dt} &= W_l(t) - D(t)P_l + \alpha(t)P_s - \beta(t)P_l, \\ \frac{dP_s}{dt} &= W_s(t) - \alpha(t)P_s + \beta(t)P_l. \end{aligned} \tag{4}$$

We note that the equations for P_l and P_s are uncoupled from that for Q. Therefore, the dynamics of (P_l, P_s) is described by

$$\begin{aligned} \frac{dP_l}{dt} &= W_l(t) - D(t)P_l + \alpha(t)P_s - \beta(t)P_l, \\ \frac{dP_s}{dt} &= W_s(t) - \alpha(t)P_s + \beta(t)P_l. \end{aligned} \tag{5}$$

It is worth noting that (5) comprises the empirical model of phosphorus recycling proposed in [26] and also the model with nutrient retention presented in [47].

Let us define $x = (P_l, P_s)^T$, then we can rewrite (5) in terms of x as follows:

$$\frac{dx}{dt} = A(t)x + b(t), \tag{6}$$

with

$$A(t) = \begin{bmatrix} -D(t) - \beta(t) & \alpha(t) \\ \beta(t) & -\alpha(t) \end{bmatrix}, \quad \text{and} \quad b(t) = \begin{bmatrix} W_l(t) \\ W_s(t) \end{bmatrix}. \tag{7}$$

Eq. (6) shows that the dynamics of phosphorus is described by a non-homogeneous linear periodic system. Applying Theorem 8 in Appendix B, we get the following result.

Proposition 1. *The system (5) admits a unique non-negative ω-periodic solution (P_l^0, P_s^0), and any solution to (5), with non-negative initial conditions, converges to it.*

Proof. Direct application of Theorem 8 in Appendix A. □

Proposition 1 states that (5) admits a unique non-negative ω-periodic solution (P_l^0, P_s^0). Furthermore, any solution to (5) with non-negative initial conditions will asymptotically approach this periodic solution. Now, using an argument based on asymptotically periodic semiflows, we can prove that any solution to (4) converges to (Q^0, P_l^0, P_s^0), with Q^0 the unique ω-periodic solution to

$$\frac{dQ}{dt} = \rho(t, Q, P_l^0(t)) - \mu(t, 0, Q)Q. \tag{8}$$

This is proved in the following proposition.

Proposition 2. *System (4) admits a unique ω-periodic solution (Q^0, P_l^0, P_s^0), and any solution to (4) with initial conditions on $\mathbb{R}_+ \times [Q^{min}, Q^{max}]$ converges to it.*

Proof. Let us define $F_0(t, Q) = \rho(t, Q, P_l^0(t))/Q - \mu(t, 0, Q)$. From Assumption 2, we have that $F_0(t, Q^{min}) \geq 0$ for all $t \geq 0$ and $\int_0^\omega F_0(t, Q^{max})dt < 0$. From Assumption 3, we have that F_0 is decreasing in Q. Moreover, from Assumption 4 we have that there is a t^* such that $F_0(t^*, u)$ is strictly decreasing in u. Thus, applying Proposition 7 Appendix A, we conclude that (8) admits a unique ω-periodic solution Q^0 satisfying $Q^0(t) \geq Q^{min}$ for all $t \geq 0$. From Proposition 1, we have that (5) admits a unique ω-periodic solution (P_l^0, P_s^0). Now, it is clear that (Q^0, P_l^0, P_s^0) is the unique ω-periodic solution to (4).

Now let $(\bar{Q}, \bar{P}_l, \bar{P}_s)$ be a solution to (4). From Proposition 1, we know that $(\bar{P}_l, \bar{P}_s)$ converges to (P_l^0, P_s^0). Therefore, it remains to prove that $\bar{Q}$ converges to Q^0. For this purpose, let us define $F(t, Q) = \rho(t, Q, \bar{P}_l(t))/Q - \mu(t, 0, Q)$. Since $\bar{P}_l$ converges to P_l^0 and ρ is uniformly Lipschitz uniformly in t, we have that $\lim_{t\to\infty} |F_0(t, Q) - F(t, Q)| = 0$ uniformly in Q for any bounded set of $\mathbb{R}_+$. Since $\bar{Q}$ is a solution of $dQ/dt = F(t, Q)Q$, from Proposition 7 (Appendix A), we conclude that $\bar{Q}$ converges to Q^0. This completes the proof. □

Remark 1. In absence of algae and time-varying parameters (autonomous case), the periodic solution provided by Proposition 1 is constant and satisfies

$$\begin{aligned} 0 &= W_l - DP_l^0 + \alpha P_s^0 - \beta P_l^0, \\ 0 &= W_s - \alpha P_s^0 + \beta P_l^0. \end{aligned}$$

from where we have that $P_0^l = (W_l + W_s)/D$, and therefore P_0^l does not depend on the values of α or β. This implies that the release and sedimentation rates have no impact on the long-term phosphorus concentration in the lake. In Section 4 (Fig. 2), we show numerically that in the presence of periodic forcing the long-term phosphorus concentration is affected by α.

3.2. Population persistence

Let (Q^0, P_l^0, P_s^0) be the unique ω-periodic solution to (4) provided by Proposition 2. We define the *basic reproduction number* for (1) as follows:

$$\mathcal{R} := \frac{\displaystyle\int_0^\omega \mu(t, 0, Q^0(t))dt}{\displaystyle\int_0^\omega [D(t) + m(t)]\, dt}. \tag{9}$$

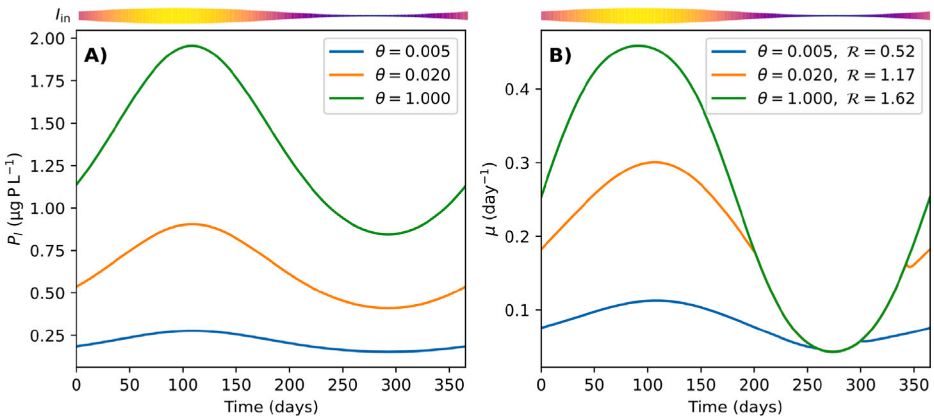

Fig. 2. Periodic regime of phosphorus concentration in the lake and invasion growth rate in absence of algae under different control levels of internal loading (θ) when P_{in} = 1 µg/l. Bars at the top represent incident light: both increased bar thickness and warmer (plasma) colors indicate higher irradiance over the seasonal cycle. **(A)** Lake Phosphorus concentration ($P_l^0(t)$) provided by Proposition 2. **(B)** Invasion specific growth rate ($\mu(t, 0, P_l^0(t))$).

Originally introduced in epidemiology [27], the concept of basic reproduction number has since been adopted in various areas of population dynamics, including chemostat theory [48,49] and metapopulation theory [50]. This number is intended to capture a fundamental threshold property: if $\mathcal{R} > 1$, the population persists, whereas if $\mathcal{R} \leq 1$, the population goes extinct. Following Chesson's coexistence theory [51], we refer to

$$\mu\big(t, 0, Q^0(t)\big)$$

as the *invasion growth rate*, that is, the long term per capita growth rate when the population is essentially absent. Hence, $\mathcal{R}$ quantifies the ratio of cumulative invasion growth to cumulative losses (dilution plus mortality) over one seasonal cycle. It thus reflects the net reproductive potential of the algal population under periodically varying environmental conditions. The following theorem, which is the main result of this work, formalizes the threshold property of $\mathcal{R}$.

Theorem 3. *Let (N, Q, P_l, P_s) be a solution to (1) with initial conditions on X defined by (3). We have that:*

(I) *If $\mathcal{R} > 1$, then algae persist uniformly in the long-term, that is, there exists $\epsilon > 0$ such that*

$$\liminf_{t\to\infty} N(t) \geq \epsilon,$$

whenever $N(0) > 0$. Moreover, the model admits an ω-periodic solution (N^*, Q^*, P_l^*, P_s^*) with $N^*(t) > 0$ for all $t \in [0, \omega]$.

(II) *If $\mathcal{R} \leq 1$, then algae go extinct, that is,*

$$\lim_{t\to\infty} N(t) = 0$$

Before proving Theorem 3, we state two lemmas regarding basic properties of (1).

Lemma 4. *We have that:*

(a) *The set X defined in (3) is positively invariant with respect to (1).*
(b) *Solutions to (1) with initial conditions in X are uniformly bounded.*

Proof. Let (N, Q, P_l, P_s) be a solution to (1) that starts within the set X. If N, P_s, or P_l becomes zero, the derivatives of these variables will be non-negative (use Assumption 1), meaning that they cannot decrease below zero. Now, consider the variable Q. According to Assumption 2, if Q reaches $Q^{\min}$, we have

$$\frac{dQ}{dt} = \rho(t, Q^{min}, P_l) \geq 0,$$

indicating that Q cannot decrease further. Conversely, if Q reaches $Q^{\max}$, we find

$$\frac{dQ}{dt} = -\mu(t, N, Q^{\max})Q^{max} \leq 0,$$

which means that Q cannot increase beyond $Q^{\max}$. This completes the proof of (a).

Now we prove that solutions to (1) are uniformly bounded. First of all, we note that

$$\begin{aligned}\frac{dP_l}{dt} &\leq W_l(t) - D(t)P_l + \alpha(t)P_s - \beta(t)P_l,\\ \frac{dP_s}{dt} &\leq W_s(t) - \alpha(t)P_s + \beta(t)P_l,\end{aligned} \tag{10}$$

Note that the right side of (10) corresponds to the right side of (5). Since (5) is a cooperative system, by a comparison argument (see Appendix B in [52]), we have that P_l and P_s are ultimately bounded by any upper bound of P_l^0 and P_s^0, respectively, with (P_l^0, P_s^0) the unique ω-periodic solution to (5) provided by Proposition 1. In particular, this shows the existence of a time $t' > 0$ and an upper bound $M > 0$ of $P_s(t)$ for all $t \geq t'$.

Now, let us define $P = NQ$ and $U = P + P_s + P_l$, then we have

$$\begin{aligned}\frac{dN}{dt} &= [\mu(t, N, Q) - D(t) - m(t)]N,\\ \frac{dP}{dt} &= \rho(t, P/N, P_l)N - (D(t) + m(t))P,\\ \frac{dU}{dt} &= W_s(t) + W_l(t) - D(t)U + D(t)P_s - m(t)P,\\ \frac{dP_s}{dt} &= W_s(t) - (\alpha(t) + \beta(t))P_s - \beta(t)P + \beta(t)U.\end{aligned} \tag{11}$$

Since $P_s(t) \leq M$ for all $t \geq t'$, from the third equation in (11) we obtain that

$$\frac{dU}{dt} \leq W_s + W_l - D(t)U + D(t)M,$$

for all $t \geq t'$. This shows that U is ultimately bounded. From the definition of U, we also conclude that P is ultimately bounded. Thus, since $Q(t) \in [Q^{min}, Q^{max}]$, and $P = NQ$ is ultimately bounded, we conclude that N is also ultimately bounded. Finally, the ultimate boundedness of solutions to (1) implies their uniform boundedness (see, Theorem 8.5 in [53]). □

Lemma 5. *If $\mathcal{R} > 1$, then any solution (N, Q, P_l, P_s) to (1) starting on X defined in (3) satisfies*

$$\limsup_{t\to\infty} N(t) > 0.$$

Proof. Let $(\bar{N}, \bar{Q}, \bar{P}_l, \bar{P}_s)$ be a solution to (1). Then $\bar{N}$ is a solution of the following ODE:

$$\frac{du}{dt} = F(t, u)u, \tag{12}$$

with $u(0) = \bar{N}(0)$ and $F(t, u) = \mu(t, u, \bar{Q}(t)) - m(t) - D(t)$. Now let us define $F_0(t, u) = \mu(t, u, Q^0(t)) - m(t) - D(t)$, with Q^0 the unique ω-periodic solution to (8). By Assumption 3, we have that F_0 and F are locally Liptschitz uniformly for $t \in [0, \omega]$. We will prove this lemma by contradiction. Let us assume that

$$\lim_{t\to\infty} \bar{N}(t) = 0. \tag{13}$$

Let us define $x(t) = (\bar{P}_l(t), \bar{P}_s(t))^T$, then x satisfies

$$\frac{dx}{dt} = A(t)x + b(t) + c(t), \tag{14}$$

with $A(t)$ and $b(t)$ given by (7), and

$$c(t) = \begin{bmatrix} -\rho(t, \bar{Q}(t), \bar{P_l}(t))\bar{N}(t) \\ 0 \end{bmatrix}.$$

From Lemma 4, we have that $\bar{Q}(t)$ and $\bar{P}_l(t)$ are ultimately bounded. Thus, using (13), we conclude that $\|c(t)\| \to 0$ as $t \to \infty$. Now, applying Lemma 11 in Appendix B to (14), we have that $(\bar{P}_l(t), \bar{P}_s(t))$ converges to $x(t) = (P_l^0, P_s^0)$ as $t \to \infty$. Following the same argument on asymptotically periodic system used in the proof of Proposition 2, we conclude that $|\bar{Q}(t) - Q^0(t)|$ converges to 0 as $t \to \infty$. Thus, we have that

$$\lim_{t\to\infty} |F(t, u) - F_0(t, u)| = 0,$$

uniformly in $t \in [0, \omega]$. We note that $\mathcal{R} > 1$ is equivalent to $\int_0^\omega F_0(t, 0)dt > 0$. Applying Theorem 6 in Appendix A, we have that any solution to (12) cannot converge to zero, which contradicts (13). Therefore $\bar{N}$ does not converge to zero, which is equivalent to $\limsup_{t\to\infty} \bar{N}(t) > 0$. □

Proof of Theorem 3 (*I*)**.** To prove the first statement, we employ Persistence Theory [30]. Let $\phi(t, u)$ denote the unique solution to (1) satisfying $\phi(0, u) = u$ with $u = (N, Q, P_l, P_s)^T$. We define the discrete semiflow $\Phi : \mathbb{Z}_+ \times X \longrightarrow X$ by $\Phi(n, u) = \phi(n\omega, u)$ with

X defined by (3). Let us define $u^0 = (0, Q_l^0(0), P_l^0(0), P_s^0(0))$, with (Q^0, P_l^0, P_s^0) the unique ω-periodic solution of (4) provided by Proposition 2. We define the persistence function $p : X \longrightarrow \mathbb{R}_+$ by $p(u) = N$. We also define the set $X_0 = X \cap (\{0\} \times \mathbb{R}_+^3)$, which comprises the elements $u \in X$ such that $p(\Phi(n, u)) = 0$ for all $n \in \mathbb{Z}_+$. Let $\Omega = \bigcup_{u \in X_0} \omega(u)$. According to Proposition 2, the ω-limit set of $u \in X_0$ is given by the singleton $M = \{u^0\}$, which implies $\Omega = M$. From Lemma 5, we have that M is weakly p-repelling, that is, there is no $u \in X$ such that $p(u) > 0$ and $\Phi(n, u) \to M$ as $n \to \infty$ (Definition 8.15 in [30]). Since $\{M\}$ is trivially acyclic (see Definition 8.14 in [30]), Theorem 8.20 in [30] guarantees that Φ is uniformly weakly p-persistent. Now, from Lemma 4, we know that there is a compact set $B \subset X$ that attracts Φ. Therefore, applying Theorem 13.9 in [30], we conclude that Φ is uniformly p-persistence.

We now define $S : X \longrightarrow X$ by $S(u) = \Phi(1, u)$. Since Φ is uniformly p-persistent, we have that S is also uniformly p-persistent. From Lemma 4 part (b), we have that S is compact. Thus, we can apply Theorem 3.1.1 by [28], and we conclude that the ω-periodic semiflow $\mathcal{T}$ defined by $\mathcal{T}(t) := \phi(t, u) : X \longrightarrow X$, with X defined by (3), is also uniformly p-persistent.

Finally, by Theorem 2.3 in [29], we conclude the existence of a non-trivial fixed point of the Poncaré map S, which is equivalent to the existence of a non-trivial ω-periodic solution to (1).

(II) Let $(\hat{N}, \hat{Q}, \hat{P}_l, \hat{P}_s)$ be any solution to (1). From the third and fourth equations of (1), we obtain:

$$\begin{aligned} \frac{d\hat{P}_l}{dt} &= W_l(t) - D(t)\hat{P}_l + \alpha(t)\hat{P}_s - \beta(t)\hat{P}_l - \rho(t, \hat{Q}, \hat{P}_l)\hat{N}, \\ \frac{d\hat{P}_s}{dt} &= W_s(t) - \alpha(t)\hat{P}_s + \beta(t)\hat{P}_l. \end{aligned}$$

By neglecting the non-positive uptake term $-\rho(t, \hat{Q}, \hat{P}_l)\hat{N}$ in the first equation, we obtain the inequality:

$$\begin{aligned} \frac{d\hat{P}_l}{dt} &\leq W_l(t) - D(t)\hat{P}_l + \alpha(t)\hat{P}_s - \beta(t)\hat{P}_l, \\ \frac{d\hat{P}_s}{dt} &= W_s(t) - \alpha(t)\hat{P}_s + \beta(t)\hat{P}_l. \end{aligned} \tag{15}$$

Let $(\bar{P}_l, \bar{P}_s)$ denote the solution to (5) with initial conditions $\bar{P}_l(0) = \hat{P}_l(0)$ and $\bar{P}_s(0) = \hat{P}_s(0)$. Since (5) is cooperative, a standard comparison argument (see Appendix B in [52]) applied to (5) and (15) implies that:

$$\hat{P}_l(t) \leq \bar{P}_l(t), \quad \hat{P}_s(t) \leq \bar{P}_s(t), \quad \forall t \geq 0. \tag{16}$$

Define $\hat{P} = \hat{N}\hat{Q}$. Then, as in the derivation of (11), we obtain:

$$\begin{aligned} \frac{d\hat{N}}{dt} &= [\mu(t, \hat{N}, \hat{P}/\hat{N}) - D(t) - m(t)]\hat{N}, \\ \frac{d\hat{P}}{dt} &= \rho(t, \hat{P}/\hat{N}, \hat{P}_l)\hat{N} - (D(t) + m(t))\hat{P}. \end{aligned}$$

Using Assumption 3, which states the monotonicity with respect to N and P_l, along with inequality (16), we obtain:

$$\begin{aligned} \frac{d\hat{N}}{dt} &\leq [\mu(t, 0, \hat{Q}) - D(t) - m(t)]\hat{N}, \\ \frac{d\hat{P}}{dt} &\leq \rho(t, \hat{Q}, \bar{P}_l)\hat{N} - (D(t) + m(t))\hat{P}. \end{aligned} \tag{17}$$

Define $(\tilde{N}, \tilde{P})$ as the unique solution to:

$$\begin{aligned} \frac{dN}{dt} &= [\mu(t, 0, P/N) - D(t) - m(t)]N, \\ \frac{dP}{dt} &= \rho(t, P/N, \bar{P}_l)N - (D(t) + m(t))P, \end{aligned} \tag{18}$$

with initial conditions $\tilde{N}(0) = \hat{N}(0)$ and $\tilde{P}(0) = \hat{P}(0)$. Using Assumption 3, we note that (18) is cooperative. Therefore, by comparison argument (see Appendix B in [52]) applied to (17) and (18), we conclude that:

$$\hat{N}(t) \leq \tilde{N}(t), \quad \hat{P}(t) \leq \tilde{P}(t), \quad \forall t \geq 0.$$

Furthermore, the variable $\tilde{Q} := \tilde{P}/\tilde{N}$ solves the following ODE:

$$\frac{dQ}{dt} = Qg(t, Q),$$

with

$$g(t, Q) = \frac{\rho(t, Q, \bar{P}_l)}{Q} - \mu(t, 0, Q).$$

Let us now define

$$g_0(t, Q) = \frac{\rho(t, Q, \bar{P}_l^0)}{Q} - \mu(t, 0, Q).$$

Table 1
Parameter definitions, units, and values for the general algal–phosphorus dynamics model (1) with rate functions from Section 2.1. Kinetic parameters are based on *Synechocystis*.

Par.	Value	Units	Description	Ref.
μ_{max}	1.27	day^{-1}	max. growth rate	[54]
Q^{min}	0.0093	pg P cell^{-1}	Subsistence quota	[18]
Q^{max}	0.2	pg P cell^{-1}	Theoretical max. quota	[18]
ν_{max}	0.46	pg P cell^{-1} d^{-1}	max. absorption rate	[18]
k	0.24×10^{-11}	m^2 cell^{-1}	Absorption coefficient	[18]
K_{bg}	7.2	m^{-1}	Background turbidity	[18]
H_I	58	μmol^{-1} m^{-2} s^{-1}	Light half-saturation	[18]
H_P	0.11	g m^{-3}	P half-saturation	[55]
D	0.006	day^{-1}	Dilution rate	[11]
m	0.15	day^{-1}	Mortality rate	[54]
T_{min}	8.3	°C	Growth min. temp.	[56]
T_{opt}	25.9	°C	Growth optimal temp.	[56]
T_{max}	35.3	°C	Growth max. temp.	[56]
b_S	0.0470	m day^{-1}	P sedimentation rate	[26]
t_S	0	–	temp. correction for b_S	[26]
b_F	0.000595	day^{-1}	P release rate	[26]
t_F	0.08	–	temp. correction for b_F	[26]
P_{in}	1–500	μg l^{-1}	P external input conc.	
L	1	m	Mean lake water depth	
T_0	18.1	°C	Temperature mean level	[57]
T_A	6.1	°C	temp. oscillation amplitude	[57]
I_0	580	μmol m^{-2} s^{-1}	Irradiance mean level	[57]
I_A	204	μmol m^{-2} s^{-1}	Irradiance oscillation amp.	[57]

Since $\bar{P}_l(t)$ converges to $P_l^0(t)$ as $t \to \infty$ (Proposition 2), we have that $\lim_{t\to\infty} |g(t,Q) - g_0(t,Q)| = 0$ uniformly for $Q \in [Q^{min}, Q^{max}]$. Moreover, $g_0(t, Q^{min}) \geq 0$ and $g(t, Q^{min}) \geq 0$ for all $t \geq 0$ and $\int_0^\omega g_0(t, Q^{max})dt < 0$. Applying Proposition 7 (Appendix A), we conclude that $\tilde{Q}(t)$ approaches asymptotically $Q^0(t)$, the unique ω-periodic solution to (8).

Finally, define $F_0(t,N) = \mu(t,N,Q^0(t)) - D(t) - m(t)$ and $F(t,N) = \mu(t,N,\tilde{Q}(t)) - D(t) - m(t)$. Since $\mathcal{R} \leq 1$ is equivalent to $\int_0^\omega F_0(t,0)dt \leq 0$, and $\tilde{N}$ solves $dN/dt = F(t,N)N$, Theorem 6 (Appendix A) implies that $\tilde{N}(t) \to 0$ as $t \to \infty$. Therefore, by comparison, $\hat{N}(t)$ also converges to zero. □

4. Numerical simulations

We conduct numerical simulations using the model parameters listed in Table 1, whose values derive from literature and consider the cyanobacteria genus *Synechocystis* as a model organism of harmful algae. Lake temperature (T) and incident light intensity (I_{in}) are defined as periodic functions with one-year period:

$$T(t) = \quad T_0 + T_A \cos\left(\frac{2\pi t}{365}\right),$$

$$I_{in}(t) = \quad I_0 + I_A \sin\left(\frac{2\pi t}{365}\right).$$

Here, T_0, I_0, T_A, and I_A represent the mean levels and oscillation amplitudes, respectively. These parameters were derived from measurements taken in [57]. To present results in terms of chlorophyll, we assumed a chlorophyll-a content of 2.16×10^{-14} g Chla/cell [58].

4.1. Reproduction number

As stated in Remark 1, when $N = 0$, in the absence of time-varying parameters, the phosphorus concentration in the lake does not depend on the release and sedimentation rates (α and β). However, with annual fluctuations in these internal loading rates, the dynamics of phosphorus concentration will vary accordingly. As Fig. 2 illustrates, reducing θ (and hence α, see (2)) lowers the equilibrium phosphorus concentration P_l^0 given by Proposition 2, thereby decreasing the algal invasion growth rate. A smaller θ corresponds to reduced sediment-derived phosphorus release, limiting nutrient availability. Consequently, the threshold parameter $\mathcal{R}$ decreases as θ is reduced.

Fig. 3 shows a heatmap with the value of the basic reproduction number $\mathcal{R}$ as a function of P_{in} and θ. We observe the existence of a region in which the ecosystem is free of algae in the long term ($\mathcal{R} < 1$). When P_{in} is higher than a critical threshold value of external phosphorus loading, denoted P_{in}^*, the persistence of algae becomes unavoidable. Conversely, when $P_{in} < P_{in}^*$, the development of the population can be avoided, especially at low θ values. It is worth noting that under no control of internal loadings ($\theta = 1$),

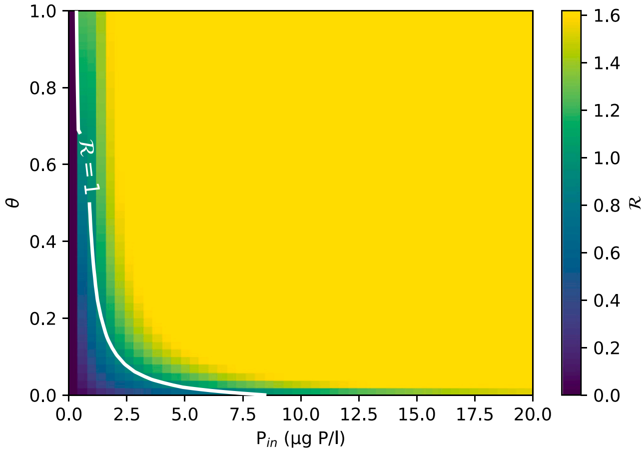

Fig. 3. Heatmap of the basic reproduction number $\mathcal{R}$ across the (P_{in}, θ) parameter space, with the contour $\mathcal{R} = 1$ delineating the boundary between algal persistence ($\mathcal{R} > 1$) and extinction ($\mathcal{R} \leq 1$).

extremely low P_{in} values are required to avoid persistent algal blooms in the long term. This resilience is explained by the capacity of the system to retain phosphorus in sediments, thus complementing the external inputs.

Fig. 3 presents a heatmap of the reproduction number $\mathcal{R}$ over the (P_{in}, θ) plane, with the contour $\mathcal{R} = 1$ delineating algal extinction ($\mathcal{R} < 1$) from persistence ($\mathcal{R} > 1$). Above a critical external loading P^*_{in}, blooms persist regardless of sediment control, whereas for $P_{\text{in}} < P^*_{\text{in}}$, blooms can be suppressed, particularly at reduced θ. Notably, when sediment release is uncontrolled ($\theta = 1$), only very low P_{in} values are sufficient to prevent persistence. This resilience arises from the sediment reservoir's ability to buffer phosphorus, sustaining algal growth even under minimal external inputs.

Note that in Fig. 2B, all curves of $\mu(t, 0, P^0_l)$ collapse between days 250 and 300, indicating that low light intensity is the sole growth constraint in this interval. This light limitation becomes even more dominant at high external phosphorus loadings. For example, at $P_{in} = 50$ µg/L and $\theta = 0$ (no internal loading), numerical simulations show that $\mu_I(t, 0) < \mu_P(Q^0(t))$ for all $t \in [0, \omega]$. Thus, the invasion growth rate reduces to $\mu(t, 0, Q^0(t)) = \phi(T(t))\mu_I(t, 0)$ (see Section 2.1). This implies that increasing θ will have no impact on $\mathcal{R}$ as any increase of it will only favor the excess of phosphorus. Indeed, as shown in Fig. 3, most of the heatmap takes the same value ($\mathcal{R} = 1.68$, yellow zone).

4.2. Control of external and internal loads in eutrophic lakes

While the value of $\mathcal{R}$ indicates whether algae survive or not in the long term, it does not provide any information about the abundance of algae, or the trophic state of the lake, in the case of algal survival. To assess the joint effects of external loading P_{in} and internal loading control θ on lake trophic state, we consider a baseline scenario with no internal control ($\theta = 1$) and $P_{\text{in}} = 500\ \mu\text{g P l}^{-1}$. In this case, the system converges to a eutrophic state, defined by an annual mean chlorophyll-a concentration above $8\ \mu\text{g l}^{-1}$. Fig. 4 shows the chlorophyll concentration along two consecutive years in this eutrophic state (time interval $[0, -2]$ in all panels). Additionally, Fig. 4 shows simulations for a period of 10 years for the transient state of the system following the onset of control measures at time $t = 0$.

As illustrated in Fig. 4, model (1) reproduces pronounced seasonal blooms similar to those generated by more complex models [14], driven by periodic variations in temperature and incident light: bloom peaks coincide with rising irradiance and temperature, while minima occur during their decline. Moreover, we observe that for different values of P_{in} and θ the solutions to (1) approach a periodic solution. Although Theorem 3 only guarantees the existence of such a periodic solution, numerical simulations suggest that this periodic solution is globally attractive.

Fig. 4 also shows the impact of different control strategies. Panel (A) illustrates that incremental reductions in P_{in} steadily diminish bloom amplitudes, with larger cuts in external loading producing correspondingly lower peak chlorophyll concentrations. In panel (B), we observe that a decrease in θ leads to a rapid decrease in bloom peaks; however, the chlorophyll concentration gradually increases to a new periodic regime. In panel (B), lowering θ produces an immediate reduction in peak chlorophyll levels, after which the system relaxes into a new, lower-amplitude periodic regime. In panel (C), simultaneous reductions of both watershed inputs and sediment release produce the largest drop in bloom peaks, immediately establishing a new, lower-amplitude periodic cycle.

To obtain a general picture on the impact of both controls, Fig. 5 presents heatmaps of the annual mean chlorophyll-a concentration following the onset of control measures, at 1, 5, and 10 years after intervention, as well as in the long-term steady state, plotted over the (P_{in}, θ) parameter plane. When P_{in} remains high and $\theta \approx 1$, the lake stays eutrophic at all horizons. Reducing P_{in} alone promotes a gradual drift toward mesotrophy, yet even after 10 years oligotrophic conditions are not achieved unless θ is also lowered. By contrast, coordinated reductions in both P_{in} and θ drive the system into the oligotrophic regime within a single year and maintain it thereafter. These results underscore that simultaneous management of external inflows and sediment-release rates is essential for rapid and sustained algal control.

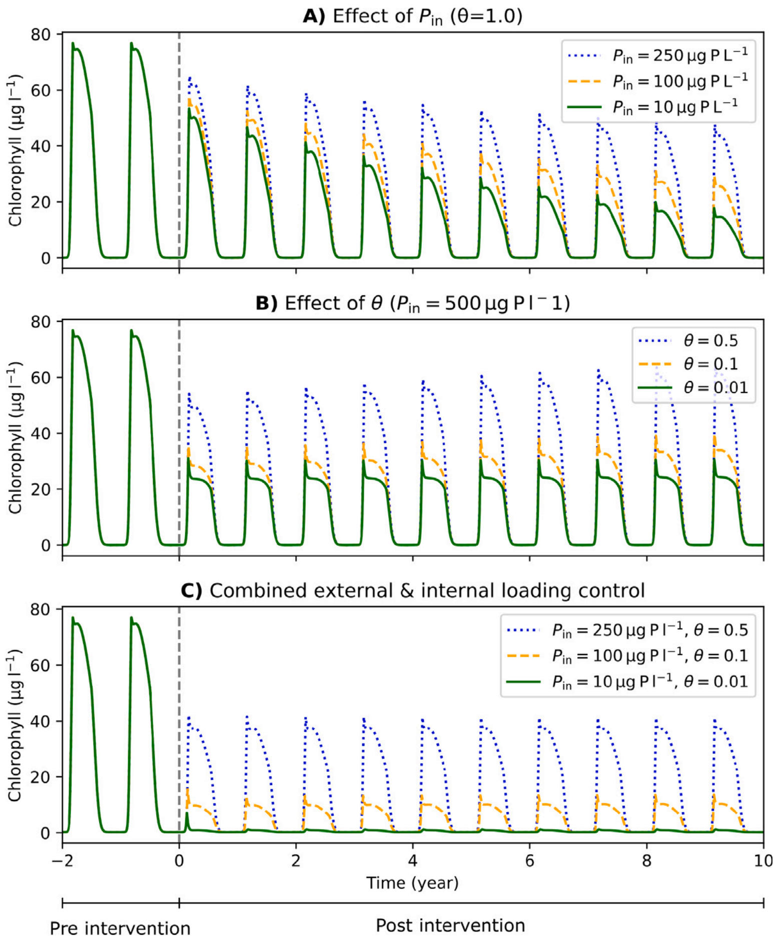

Fig. 4. Time-series of chlorophyll-a following phosphorus controls (2 years baseline, control from $t = 0$): **(A)** Progressive reductions in external loading P_{in} steadily lower bloom amplitudes; **(B)** Decreasing the internal release parameter θ produces an immediate drop in peak concentrations, then settles into a lower-amplitude cycle; **(C)** Simultaneous reductions in both P_{in} and θ instantly suppress blooms and establish a stable, low-amplitude periodic regime.

5. Conclusions

5.1. Mathematical analysis

We have developed and analyzed a periodically forced ODE model that captures harmful algal growth in shallow lakes with both external and sediment-driven internal phosphorus loadings. We identified necessary and sufficient conditions for the persistence of algae through the definition of a basic reproduction number ($\mathcal{R}$), which determines the persistence or extinction of the population (Theorem 3). To prove such a result, we have firstly established results on phosphorus dynamics in the absence of algae. In several single population models, the critical condition for persistence is determined by evaluating the growth rate at the nutrient concentration that the system reaches in the absence of the population, which dynamics are generally given by a one-dimensional ODE. However, for the model studied here, the dynamics of nutrients is described by a two-dimensional non-homogeneous linear system with periodic coefficients. Such a kind of system may present complex behaviors such as multiple periodic solutions [60]. We have applied Lyapunov type results to prove the global asymptotic stability of the origin for the homogeneous linear component, and the Floquet theory to prove the uniqueness of a globally asymptotically stable periodic solution. It is important to note that such results represent also a contribution in the field of linear non-autonomous systems. Based on these results, we have applied results from persistence theory and periodic semiflows to conclude the persistence of algae.

5.2. Management strategies

From a practical point of view, we have proposed a minimal conceptual model that elucidates how internal nutrient recycling drives long-term phytoplankton blooms in shallow lakes. We showed that when release and sedimentation rates are constant, these internal fluxes have no impact on the basic reproduction number. However, under seasonal forcing (e.g. temperature-driven

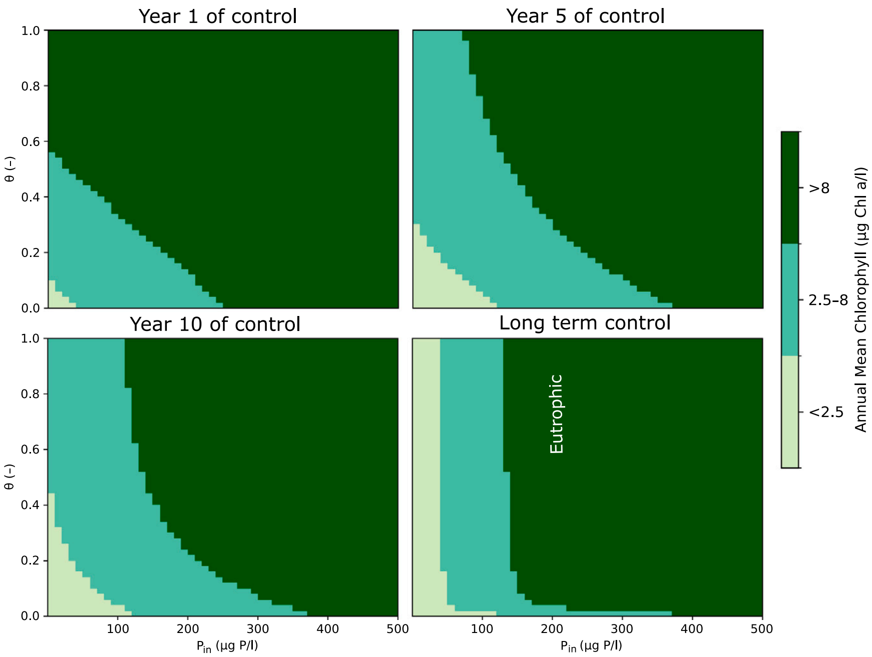

Fig. 5. Heatmaps of the annual mean chlorophyll-a concentration during the final year of 1, 5, and 10 years after intervention, and in the long-term steady state (defined by $|N(t+\omega) - N(t)| < 0.01\,|N(t)|$ with $\omega = 1$ year), plotted over the (P_{in}, θ) parameter plane. Trophic categories (oligotrophic, mesotrophic, eutrophic) are annotated based on chlorophyll thresholds from [59].

variation), modulating those fluxes can prevent persistent blooms, but only when external phosphorus inputs are kept extremely low, conditions probably difficult to meet in most lakes. While completely avoiding the persistence of algae is very difficult, simulations show that simultaneously managing both external and internal phosphorus inputs can effectively reduce chlorophyll concentrations (see Fig. 5). Indeed, numerical simulations show that even at very low external phosphorus loading, internal phosphorus supply can maintain cyclic blooms in a mesotrophic lake during several years, whereas a combined reduction of both external and internal inputs markedly accelerates recovery, underscoring the necessity of targeting internal loading in restoration strategies.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to enhance clarity of some sentences. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. One-dimensional periodic Kolmogorov equation

We present two results in this section. The first result corresponds to a weaker version of Theorem 2.1 by [36] for asymptotically periodic Kolmogorov equations. Consider the non-autonomous Kolmogorov equation:

$$\frac{du}{dt} = uF(t,u), \quad u \in \mathbb{R}^+ = [0,\infty), \tag{A.1}$$

and the ω-periodic Kolmogorov equation:

$$\frac{du}{dt} = uF_0(t,u), \quad u \in \mathbb{R}^+, \tag{A.2}$$

where $F(t,u) : \mathbb{R}^2_+ \to \mathbb{R}$ is continuous, decreasing in u, and locally Lipschitz in u, and $F_0(t,u) : \mathbb{R}^2_+ \to \mathbb{R}$ is continuous, ω-periodic in t $(\omega > 0)$, decreasing in u, and locally Lipschitz in u uniformly in $t \in [0,\omega]$. Consider the following assumptions:

Assumption A.1. *$\lim_{t\to\infty} |F(t,u) - F_0(t,u)| = 0$ uniformly for u in any bounded subset of $\mathbb{R}^+$.*

Assumption A.2. *$\int_0^\omega F_0(t,R)\,dt < 0$ for some $R > 0$.*

Assumption A.3. *There exists $t' \geq 0$ such that the function $u \mapsto F_0(t',u)$ is strictly decreasing.*

Let us define $\phi(t,u)$ as the unique solution to (A.1) such that $\phi(0,u) = u$. The main difference between the next result and Theorem 2.1 proposed by [36] is the relaxation of the restriction on F, which is now allowed to be decreasing without strictness. The proof is rather similar to that presented by [36].

Theorem 6. *Assume that Assumptions A.1–A.3 hold. We have*

(I) *If $\int_0^\omega F_0(t,0)\,dt \leq 0$, then for any $u \in \mathbb{R}^+$, $\lim_{t\to\infty} \phi(t,0,u) = 0$;*
(II) *If $\int_0^\omega F_0(t,0)\,dt > 0$, then for any $u \in \mathbb{R}^+ \setminus \{0\}$, $\lim_{t\to\infty}(\phi(t,u) - u^*(t)) = 0$, where $u^*(t)$ is the unique positive ω-periodic solution of the periodic Kolmogorov equation $\frac{du}{dt} = uF_0(t,u)$.*

Proof. Let $\phi_0(t,u)$, $t \geq 0$ be the unique solution of the ω-periodic Eq. (A.2). We claim that

(a) If $\int_0^\infty F_0(t,0)\,dt \leq 0$, then for any $u \in \mathbb{R}^+$, $\lim_{t\to\infty} \phi_0(t,0,u) = 0$;
(b) If $\int_0^\infty F_0(t,0)\,dt > 0$, then (A.2) admits a unique ω-periodic solution u^* and for any $u \in \mathbb{R}^+$, $\lim_{t\to\infty}(\phi(t,0,u) - u^*(t)) = 0$. then for any $u \in \mathbb{R}^+ \setminus \{0\}$, $\lim_{t\to\infty}(\phi_0(t,u) - u^*(t)) = 0$, where $u^*(t)$ is the unique positive ω-periodic solution of the periodic Kolmogorov equation $\frac{du}{dt} = uF_0(t,u)$.

From Lemma 3.3 by [61], we have that (a) holds when the inequality is strict. Now, since F_0 is decreasing in u and strictly decreasing in u for some $t \in [0,\omega]$, we have that

$$\int_0^\omega F_0(t,u)dt < \int_0^\omega F_0(t,0)dt = 0.$$

Therefore, the Poincaré map $Q : \mathbb{R}_+ \to \mathbb{R}_+$ defined by $Q(u) = \phi_0(\omega,u)$ satisfies

$$Q(u) = u\exp\left(\int_0^\omega F(t,u)dt\right) < u.$$

This implies that Q admits no positive fixed point and that $Q^{n+1}(u) < Q^n(u)$ for all $n \geq 0$. Thus, $Q^n(u)$ necessarily converges to 0 as $n \to \infty$, hence $\phi_0(t,u)$ approaches 0 asymptotically. Thus, (a) is proved.

From Lemma A.3 by [22], we have that the solutions of (A.1) and (A.2) are ultimately bounded. Applying Lemma 1 by [36], we have the existence of $\delta > 0$ such that $\limsup_{t\to\infty} \phi_0(t,u) \geq \delta$ for all $u > 0$. Thus, (A.2) is permanent. Applying Lemma 2 by [62], there exist a minimal and a maximal positive ω-periodic solutions, denoted respectively by $u_m(\cdot)$ and $u_M(\cdot)$ to (A.2) such that

$$\liminf_{t\to\omega}(\phi_0(t,u) - u_m(t)) = 0 \quad \text{and} \quad \limsup_{t\to\omega}(\phi_0(t,u) - u_M(t)) = 0,$$

for any $u > 0$. Now, if u_m is different from u_M, we have that $u_m(t) < u_M(t)$ for all $t \in [0,\omega]$. Since F_0 is strictly decreasing for some t, we have that

$$\int_0^\omega F_0(t,u_m(t))dt < \int_0^\omega F_0(t,u_M(t))dt. \tag{A.3}$$

However, since u_m and u_M are ω-periodic solutions to (A.2), we have that

$$\int_0^\omega F_0(t,u_m(t))dt = \int_0^\omega F_0(t,u_M(t))dt = 0,$$

which contradicts (A.3). Therefore, $u^*(t) = u_m(t) = u_M(t)$ and (b) is proved.

The completion of the proof follows from extending the results of the claim to the asymptotically Kolmogorov Eq. (A.1). This extension is achieved by employing the Theory of Asymptotically Periodic Semiflows [63], and the arguments are the same as in the proof of Theorem 2.1 by [36]. □

The second result is a variation of Proposition A.4 in [22].

Proposition 7. *Assume that Assumptions A.1–A.3 hold. Let $a > 0$ and $J = [a,\infty)$, and let us assume that $F_0(t,a) \geq 0$ and $F(t,a) \geq 0$ for all $t \geq 0$. Then (A.2) admits a unique ω-periodic solution u^* and for any $u \in \mathbb{R}^+$, $\lim_{t\to\infty}(\phi(t,0,u) - u^*(t)) = 0$.*

Proof. We apply Proposition A.4 part (a) from [22] to obtain that (A.2) admits at least one ω-period solution. Based on Assumption A.3, using exactly the same arguments as in the proof of Theorem 8, we can prove that this ω-periodic solution is unique. Now, applying Proposition A.4 part (b) from [22] we complete the proof. □

Appendix B. Two dimensional non-homogeneous linear periodic system

We study the following non-homogeneous linear system with periodic coefficients. The system is given by

$$\frac{dx}{dt} = A(t)x + b(t), \tag{B.1}$$

where $x \in \mathbb{R}^2_+$, $A(t)$ is a 2×2 matrix with periodic coefficients of period $\omega > 0$, and $b(t)$ is a periodic vector function with the same period as $A(t)$. Equivalently, in component form:

$$\begin{cases} \dfrac{dx_1}{dt} = a_{11}(t)\,x_1 + a_{12}(t)\,x_2 + b_1(t), \\ \dfrac{dx_2}{dt} = a_{21}(t)\,x_1 + a_{22}(t)\,x_2 + b_2(t). \end{cases}$$

We assume that

Assumption B.1. *The coefficients $a_{21}(t), a_{12}(t), b_1(t)$, and $b_2(t)$ are continuous and non-negative for all $t \in [0, \omega]$.*

Assumption B.2. *$d(t) := -[a_{11}(t) + a_{21}(t)] > 0$ and $a_{22}(t) + a_{12}(t) = 0$ for all $t \in [0, \omega]$.*

Theorem 8. *If Assumptions B.1 and B.2 hold, then Equation (B.1) admits a unique non-negative ω-periodic solution. Moreover, any solution with non-negative initial conditions converges to it.*

The proof of Theorem 8 centers around the Floquet Theory [39] and Lyapunov-type stability results from nonlinear control theory [64]. The proof is structured around the two following lemmas.

Lemma 9. *Assume that Assumption B.1 holds. Then the set $\mathbb{R}^2_+$ is positively invariant with respect to (B.1).*

Proof. Let us define the variable $y = x_1 x_2$. Then we have

$$\frac{dy}{dt} = Tr(A)y + a_{21}x_1^2 + a_{12}x_2^2,$$

with $Tr(A) = a_{11} + a_{22}$ the trace of A. Since $a_{21}(t)$ and $a_{12}(t)$ are non-negative, we have that

$$\frac{dy}{dt} \geq Tr(A)y,$$

which implies that

$$y(t) \geq y(0)exp\left(-\int_0^{\omega} Tr(A(s))ds\right) \geq 0.$$

This implies that for any $t \geq 0$, $x(t) = (x_1(t), x_2(t))$ is in the first or third quadrant. Now, if x moves to the third quadrant from the first quadrant, necessarily there is t_0 such that $x(t_0) = 0$. Then, we have that

$$\frac{dx(t_0)}{dt} = b(t_0) \geq 0.$$

Therefore, no component of x can become negative. Thus, x remains in the first quadrant and the proof is complete. □

Lemma 10. *Assume that Assumptions B.1 and B.2 hold. If $b_1 \equiv b_2 \equiv 0$, then the origin is asymptotically stable with respect to (B.1).*

Proof. Set $D = \mathbb{R}^2_+$ and let $V : \mathbb{R}_+ \times D \to \mathbb{R}$ be the continuously differentiable function defined by

$$V(t, x) = px_1^2 + 2px_1x_2 + qx_2^2,$$

with $p < q$ positive real numbers. We can easily verify that

$$p\|x\|^2 \leq V(t, x) \leq 2q\|x\|^2, \quad \text{for all } x \in D,$$

with $\|\cdot\|$ the Euclidean norm. Following Theorem 4.10 in [64], we have to prove the existence of a positive constant k_3 such that

$$\frac{\mathrm{d}V}{\mathrm{d}t} \leq -k_3\|x\|^2, \quad \text{for all } x \in D. \tag{B.2}$$

We have that

$$\frac{dV}{dt} = 2p(a_{11} + a_{21})x_1^2 + 2(pa_{12} + qa_{22})x_2^2 + 2[p(a_{12} + a_{22} + a_{11}) + qa_{21}]x_1x_2$$

Using Assumption B.2, we have:

$$\frac{dV}{dt} = -2pdx_1^2 - 2a_{12}(q - p)x_2^2 + 2[pa_{11} + qa_{21}]x_1x_2.$$

Since $a_{21} + a_{11} < 0$ and $a_{21} \geq 0$, we have that $a_{11} < -a_{21} \leq 0$ and that $-\frac{a_{21}}{a_{11}} < 1$. Now let us define

$$a_0 = \max_{t\in[0,\omega]} -\frac{a_{21}(t)}{a_{11}(t)}.$$

Since $a_0 \in (0,1)$, we can choose p and q such that they satisfy:

$$a_0 < \frac{p}{q} < 1.$$

This implies that $\delta := q - p > 0$ and $pa_{11} + qa_{21} < 0$. Let us define $\gamma := \min_{t\in[0,\omega]}\{pd(t), \delta a_{12}(t)\}$ and let $\beta \in (0,1)$, then we have:

$$\frac{dV}{dt} \leq -2pdx_1^2 - 2a_{12}\delta x_2^2 \leq -\gamma\|x\|^2.$$

Applying Theorem 4.10 from [64], we conclude that the origin is globally exponentially stable. □

Proof of Theorem 8. By Lemma 10, all the characteristic (Floquet) multipliers of the ω-periodic homogeneous system

$$\frac{dx}{dt} = A(t)x$$

lie strictly inside the unit circle. Hence, by Theorem 2.118 in [39], the non-homogeneous system (B.1) admits at least one ω-periodic solution, which we denote $y^*(t)$. Now let $y(t)$ be any solution of (B.1), and set

$$z(t) = y(t) - y^*(t).$$

Then

$$\frac{dz}{dt} = A(t)\,y + b(t) - \left[A(t)\,y^* + b(t)\right] = A(t)\,z,$$

meaning that $z(t)$ satisfies the homogeneous equation. Applying Lemma 10 once more gives

$$z(t) \longrightarrow 0 \quad \text{as } t \to \infty,$$

i.e. $y(t) \to y^*(t)$. In particular, if there were two distinct ω-periodic solutions $y_1(t)$ and $y_2(t)$, then their difference $z(t) = y_1(t) - y_2(t)$ would both be periodic and tend to zero as $t \to \infty$, forcing $z \equiv 0$ and hence $y_1 \equiv y_2$. This proves the uniqueness of the periodic solution and that every solution converges to it.

Finally, since $\mathbb{R}^2_+$ is positively invariant for (B.1), any trajectory starting in $\mathbb{R}^2_+$ remains there and thus converges to the unique ω-periodic solution. It follows that $y^*(t) \in \mathbb{R}^2_+$ for all $t \in [0,\infty)$. □

We end this section with the following asymptotic result.

Lemma 11. *Consider the system*

$$\frac{dx}{dt} = A(t)x + b(t) + c(t). \tag{B.3}$$

If $\|c(t)\| \to 0$ as $t \to \infty$, then the solutions to (B.3) converge to the unique non-negative ω-periodic solution to (B.1) provided by Theorem 8.

Proof. Let us define $F_0(t,x) = A(t)x + b(t)$ and $F(t,x) = A(t)x + b(t) + c(t)$, and let x^* be the unique non-negative ω-periodic solution to (B.1). It is clear that $\|F_0(t,x) - F_0(t,x)\| = \|c(t)\| \to 0$ as $t \to \infty$ and that solutions to (B.1) and (B.1) are uniformly bounded. Let $\phi(t,s,u)$ be the unique solution to (B.3) with $\phi(s,s,u) = u$ and let $\phi_0(t,s,u)$ be the unique solution to (B.1) with $\phi_0(s,s,u) = u$. By Proposition 3.2 in [63], ϕ is asymptotic to the ω-periodic semiflow $\mathcal{T}(t) := \phi_0(t,0,\cdot) : \mathbb{R}^+ \to \mathbb{R}^+$, and hence $\mathcal{T}^n(u) = \phi(n\omega, 0, u)$, $n \geq 0$, is an asymptotically autonomous discrete dynamical process with limit discrete semiflow $S^n : \mathbb{R}^+ \to \mathbb{R}^+, n \geq 0$. Since $x^*(0)$ is the unique globally stable fixed point of S, by Theorem 2.4 in [63], we conclude that $\lim_{n\to\infty} \mathcal{T}^n(u) = u^*(0)$ for any $u \in J$. Applying Theorem 3.1 in [63], we conclude the proof. □

Data availability

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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